

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Busulfan Kabi Injection 6mg/ml Concentrate for Solution for Infusion

WARNING MYELOSUPPRESSION

Busulfan Injection causes severe and prolonged myelosuppression at the recommended dosage. Hematopoietic progenitor cell transplantation is required to prevent potentially fatal complications of the prolonged myelosuppression.

DESCRIPTION:

Busulfan Kabi Injection 6mg/ml Concentrate for Solution for Infusion is intended for intravenous administration. It is supplied as a clear, colorless, viscous solution free from visible particles.

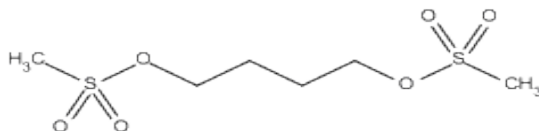
COMPOSITION:

Each ml Contains:

Busulfan USP	6 mg
N, N Dimethylacetamide Ph.Eur.	0.33ml
Polyethylene glycol USNF	0.67ml

CHEMICAL STRUCTURE:

The chemical name for Busulfan, USP is 1,4-butanediol, dimethanesulfonate. The molecular formula of Busulfan, USP is $C_6H_{14}O_6S_2$ (molecular weight- 246.30) and the structure is:



PHARMACOLOGY:

Pharmacodynamics:

Busulfan is a bifunctional alkylating agent in which two labile methanesulfonate groups are attached to opposite ends of a four-carbon alkyl chain. In aqueous media, Busulfan hydrolyzes to release the methanesulfonate groups. This produces reactive carbonium ions that can alkylate DNA. DNA damage is thought to be responsible for much of the cytotoxicity of Busulfan.

Pharmacokinetics:

Pharmacokinetics in adults

Absorption

The pharmacokinetics of intravenous Busulfan was studied in 124 evaluable patients following a 2-hour intravenous infusion for a total of 16 doses over four days. Immediate and complete availability of the dose is obtained after intravenous infusion of Busulfan. Similar blood exposure was observed when comparing plasma concentrations in adult patients receiving oral

and intravenous Busulfan at 1 mg/kg and 0.8 mg/kg respectively. Low inter (CV=21%) and intra (CV=12%) patient variability on Busulfan exposure was demonstrated through a population pharmacokinetic analysis, performed on 102 patients.

Distribution

Terminal volume of distribution V_z ranged between 0.62 and 0.85 l/kg.

Busulfan concentrations in the cerebrospinal fluid are comparable to those in plasma although these concentrations are probably insufficient for anti-neoplastic activity.

Reversible binding to plasma proteins was around 7% while irreversible binding, primarily to albumin, was about 32%.

Biotransformation

Busulfan is metabolised mainly through conjugation with glutathione (spontaneous and glutathione-S-transferase mediated). The glutathione conjugate is then further metabolised in the liver by oxidation. None of the metabolites is thought to contribute significantly to either efficacy or toxicity.

Elimination

Total clearance in plasma ranged 2.25 - 2.74 ml/minute/kg. The terminal half-life ranged from 2.8 to 3.9 hours.

Approximately 30% of the administered dose is excreted into the urine over 48 hours with 1% as unchanged Busulfan. Elimination in faeces is negligible. Irreversible protein binding may explain the incomplete recovery. Contribution of long-lasting metabolites is not excluded.

Special populations

Hepatic or renal impairment

The effects of renal dysfunction on intravenous Busulfan disposition have not been assessed.

The effects of hepatic dysfunction on intravenous Busulfan disposition have not been assessed. Nevertheless the risk of liver toxicity may be increased in this population.

No age effect on Busulfan clearance was evidenced from available intravenous Busulfan data in patients over 60 years.

INDICATIONS:

Busulfan Injection is indicated for use in combination with cyclophosphamide as a conditioning regimen prior to allogeneic hematopoietic progenitor cell transplantation for chronic myelogenous leukemia.

CONTRAINDICATIONS:

Busulfan is contraindicated in patients with a history of hypersensitivity to any of its components.

ADVERSE EFFECTS:

Tabulated summaries of adverse reactions

Frequencies are defined as: very common ($\geq 1/10$), common ($\geq 1/100$, $< 1/10$), uncommon ($\geq 1/1,000$, $< 1/100$) or not known (cannot be estimated from the available data). Undesirable effects coming from post-marketing survey have been implemented in the tables with the incidence "not known".

Busulfan in combination with cyclophosphamide or melphalan

Adverse reactions reported both in adults and paediatric patients as more than an isolated case are listed below, by system organ class and by frequency. Within each frequency grouping, adverse events are presented in order of decreasing seriousness.

System organ class	Very common	Common	Uncommon	Not known
Infections and infestations	Rhinitis Pharyngitis			
Blood and lymphatic system disorders	Neutropenia Thrombocytopenia Febrile neutropenia Anaemia Pancytopenia			
Immune system disorders	Allergic reaction			
Endocrine disorders				Hypogonadism**
Metabolism and nutrition disorders	Anorexia Hyperglycaemia Hypocalcaemia Hypokalaemia Hypomagnesaemia Hypophosphatemia	Hyponatraemia		
Psychiatric disorders	Anxiety Depression Insomnia	Confusion	Delirium Nervousness Hallucination Agitation	
Nervous system disorders	Headache Dizziness		Seizure Encephalopathy Cerebral haemorrhage	
Eye disorders				Cataract Corneal thinning Lens disorders***
Cardiac-disorders	Tachycardia	Arrhythmia Atrial fibrillation Cardiomegaly Pericardial effusion Pericarditis	Ventricular Extrasystoles Bradycardia	
Vascular disorders	Hypertension Hypotension Thrombosis Vasodilatation		Femoral artery thrombosis Capillary leak syndrome	

System organ class	Very common	Common	Uncommon	Not known
Respiratory thoracic and mediastinal disorders	Dyspnoea Epistaxis Cough Hiccup	Hyperventilation Respiratory failure Alveolar haemorrhages Asthma Atelectasis Pleural effusion	Hypoxia	Interstitial lung disease**
Gastrointestinal disorders	Stomatitis Diarrhoea Abdominal pain Nausea Vomiting Dyspepsia Ascites Constipation Anus discomfort	Haematemesis Ileus Oesophagitis	Gastrointestinal haemorrhage	Tooth hypoplasia**
Hepato-biliary disorders	Hepatomegaly Jaundice	Veno occlusive liver disease *		
Skin and subcutaneous tissue disorders	Rash Pruritis Alopecia	Skin desquamation Erythema Pigmentation disorder		
Musculoskeletal and connective tissue disorders	Myalgia Back pain Arthralgia			
Renal and urinary disorders	Dysuria Oligurea	Haematuria Moderate renal insufficiency		
Reproductive system and breast disorders				Premature menopause Ovarian failure**
General disorders and administration site conditions	Asthenia Chills Fever Chest pain Oedema Oedema general Pain Pain or inflammation at injection site Mucositis			

System organ class	Very common	Common	Uncommon	Not known
Investigations	Transaminases increased Bilirubin increased GGT increased Alkaline phosphatases increased Weight increased Abnormal breath sounds Creatinine elevated	Bun increase Decrease ejection fraction		

*veno occlusive liver disease is more frequent in paediatric population.

** reported in post marketing with IV Busulfan

*** reported in post marketing with oral Busulfan

Busulfan in combination with fludarabine (FB)

The incidence of each adverse reactions presented in the following table has been defined according to the highest incidence observed in published clinical trials in RIC regimen for which the population treated with FB was clearly identified, whatever the schedules of Busulfan administrations and endpoints. Adverse reactions reported as more than an isolated case are listed below, by system organ class and by frequency.

System organ class	Very common	Common	Not known*
Infections and infestations	Viral infection CMV reactivation EBV reactivation Bacterial infection	Invasive fungal infection Pulmonary infection	Brain abscess Cellulitis Sepsis
Blood and lymphatic system disorders			Febrile neutropenia
Metabolism and nutrition disorders	Hypoalbuminaemia Electrolyte disturbance Hyperglycaemia		Anorexia
Psychiatric disorders			Agitation Confusional state Hallucination
Nervous system disorders		Headache Nervous system disorders [Not Elsewhere Classified]	Cerebral haemorrhage Encephalopathy
Cardiac disorders			Atrial fibrillation
Vascular disorders		Hypertension	
Respiratory thoracic and mediastinal disorders		Pulmonary haemorrhage	Respiratory failure
Gastro-intestinal disorders	Nausea Vomiting Diarrhoea Stomatitis		Gastro-intestinal haemorrhage Tooth hypoplasia*
Hepato-biliary disorders	Veno occlusive liver disease		Jaundice Liver disorders

Skin and subcutaneous tissue disorders		Rash	
Renal and urinary disorders	Haemorrhagic cystitis**	Renal disorder	Oliguria
General disorders and administration site conditions	Mucositis		Asthenia Oedema Pain
Investigations	Transaminases increased Bilirubine increased Alkaline phosphatases increased	Creatinine elevated	Blood lactate dehydrogenase increased Blood uric acid increased Blood urea increased GGT increased Weight increased

* reported in post marketing experience

** include haemorrhagic cystitis induced by viral infection

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via national procedures.

DRUG INTERACTIONS:

Giving *Itraconazole* with Busulfan resulted in a decrease in the clearance of Busulfan; *Fluconazole* had no such effect. Busulfan doses may need to be decreased if *Itraconazole* is also given.

Phenytoin increases the clearance of Busulfan, possibly due to the induction of glutathione-S-transferase. The clearance of Busulfan at the recommended dose may be lower and exposure (AUC) higher in patients not treated with phenytoin.

Because Busulfan is eliminated from the body via conjugation with glutathione, use of acetaminophen prior to (< 72 hours) or concurrent with Busulfan may result in reduced Busulfan clearance based upon the known property of acetaminophen to decrease glutathione levels in the blood and tissues.

Metronidazole increase plasma levels of Busulfan, which may lead to treatment-related toxicities.

PRECAUTIONS AND WARNINGS:

The consequence of treatment with Busulfan at the recommended dose and schedule is profound myelosuppression, occurring in all patients. Severe granulocytopenia, thrombocytopenia, anaemia, or any combination thereof may develop. Frequent complete blood counts, including differential white blood cell counts, and platelet counts should be monitored during the treatment and until recovery is achieved.

Prophylactic or empiric use of anti-infectives (bacterial, fungal, viral) should be considered for the prevention and management of infections during the neutropenic period. Platelet and red

blood cell support, as well as the use of growth factors such as granulocyte colony stimulating agent (G-CSF), should be employed as medically indicated.

Hepatic impairment

Busulfan has not been studied in patients with hepatic impairment. Since Busulfan is mainly metabolized through the liver, caution should be observed when Busulfan is used in patients with pre-existing impairment of liver function, especially in those with severe hepatic impairment.

It is recommended when treating these patients that serum transaminase, alkaline phosphatase, and bilirubin should be monitored regularly 28 days following transplant for early detection of hepatotoxicity.

Hepatic veno-occlusive disease is a major complication that can occur during treatment with Busulfan. Patients who have received prior radiation therapy, greater than or equal to three cycles of chemotherapy, or prior progenitor cell transplant may be at an increased risk.

Caution should be exercised when using paracetamol prior to (less than 72 hours) or concurrently with Busulfan due to a possible decrease in the metabolism of Busulfan.

As documented in clinical studies, no treated patients experienced cardiac tamponade or other specific cardiac toxicities related to Busulfan. However cardiac function should be monitored regularly in patients receiving Busulfan.

Occurrence of acute respiratory distress syndrome with subsequent respiratory failure associated with interstitial pulmonary fibrosis was reported in Busulfan studies in one patient who died, although, no clear aetiology was identified. In addition, Busulfan might induce pulmonary toxicity that may be additive to the effects produced by other cytotoxic agents. Therefore, attention should be paid to this pulmonary issue in patients with prior history of mediastinal or pulmonary radiation.

Periodic monitoring of renal function should be considered during therapy with Busulfan.

Seizures have been reported with high dose Busulfan treatment. Special caution should be exercised when administering the recommended dose of Busulfan to patients with a history of seizures. Patients should receive adequate anticonvulsant prophylaxis.

The increased risk of a second malignancy should be explained to the patient. On the basis of human data, Busulfan has been classified by the International Agency for Research on Cancer (IARC) as a human carcinogen. The World Health Organisation has concluded that there is a causal relationship between Busulfan exposure and cancer. Leukaemia patients treated with Busulfan developed many different cytological abnormalities, and some developed carcinomas. Busulfan is thought to be leukemogenic.

Pregnancy

Haematopoietic progenitor cell transplantation (HPCT) is contraindicated in pregnant women; therefore, Busulfan is contraindicated during pregnancy. Studies in animals have shown reproductive toxicity (embryo-foetal lethality and malformations).

There are no or limited amount of data from the use of Busulfan or the solvent, Dimethylacetamide (DMA), in pregnant women. A few cases of congenital abnormalities have

been reported with low-dose oral Busulfan, not necessarily attributable to the active substance, and third trimester exposure may be associated with impaired intrauterine growth. Women of childbearing potential have to use effective contraception during and up to 6 months after treatment.

Breast-feeding

It is unknown whether Busulfan and DMA are excreted in human milk. Because of the potential for tumorigenicity shown for Busulfan in human and animal studies, breast-feeding should be discontinued during treatment with Busulfan.

Fertility

Busulfan and DMA can impair fertility in man or woman. Therefore it is advised not to father child during the treatment and up to 6 months after treatment and to seek advice on cryo-conservation of sperm prior to treatment because of the possibility of irreversible infertility.

DOSAGE AND ADMINISTRATION:

Administer Busulfan in combination with cyclophosphamide as a conditioning regimen prior to bone marrow or peripheral blood progenitor cell replacement. For adult patients, the recommended doses (BuCY2 regimen) are:

- Busulfan 0.8 mg/kg (ideal body weight or actual body weight, whichever is lower) intravenously via a central venous catheter as a two-hour infusion every six hours for four consecutive days for a total of 16 doses (Days -7, -6, -5 and -4).

- Cyclophosphamide 60 mg/kg intravenously as a one-hour infusion on each of two days beginning no sooner than six hours following the 16th dose of Busulfan (Days -3 and -2).

- Administer hematopoietic progenitor cells on Day 0.

- Premedicate patients with anticonvulsants (e.g., benzodiazepines, phenytoin, valproic acid or levetiracetam) to prevent seizures reported with the use of high dose Busulfan. Administer anticonvulsants 12 hours prior to Busulfan to 24 hours after the last dose of Busulfan.

- Administer antiemetics prior to the first dose of Busulfan and continue on a fixed schedule through Busulfan administration.

- Busulfan clearance is best predicted when the Busulfan dose is administered based on adjusted ideal body weight. Dosing Busulfan based on actual body weight, ideal body weight or other factors can produce significant differences in Busulfan clearance among lean, normal and obese patients.

- Calculate ideal body weight (IBW) as follows (height in cm, and weight in kg):

Men: $IBW (kg) = 50 + 0.91 \times (\text{height in cm} - 152)$

Women: $IBW (kg) = 45 + 0.91 \times (\text{height in cm} - 152)$

- For obese or severely obese patients, base Busulfan dosing on adjusted ideal body weight (AIBW):

$AIBW = IBW + 0.25 \times (\text{actual weight} - IBW).$

Pediatrics: The effectiveness of Busulfan in the treatment of CML has not been specifically studied in paediatrics patients.

Method of administration

Precautions to be taken before handling or administering the medicinal product

This medicinal product must be diluted prior to administration. A final concentration of approximately 0.5 mg/ml Busulfan should be achieved. Busulfan should be administered by intravenous infusion via central venous catheter.

Busulfan should not be given by rapid intravenous, *bolus* or peripheral injection.

All patients should be pre-medicated with anticonvulsant medicinal products to prevent seizures reported with the use of high dose Busulfan.

It is recommended to administer anticonvulsants 12 h prior to Busulfan to 24 h after the last dose of Busulfan.

Antiemetics should be administered prior to the first dose of Busulfan and continued on a fixed schedule according to local practice through its administration.

Preparation and Administration Precautions

Preparation of Busulfan

Procedures for proper handling and disposal of anticancer medicinal products should be considered.

All transfer procedures require strict adherence to aseptic techniques, preferably employing a vertical laminar flow safety hood.

As with other cytotoxic compounds, caution should be exercised in handling and preparing the Busulfan solution:

- The use of gloves and protective clothing is recommended.
- If the concentrate or diluted Busulfan solution contacts the skin or mucosa, wash them thoroughly with water immediately.

Calculation of the quantity of Busulfan to be diluted and of the diluent

Busulfan must be diluted prior to use with either sodium chloride 9 mg/ml (0.9%) solution for injection or glucose solution for injection 5%.

The quantity of the diluent must be 10 times the volume of the Busulfan ensuring the final concentration of Busulfan remains at approximately 0.5 mg/ml. By example:

The amount of Busulfan and diluent to be administered would be calculated as follows:
for a patient with a Y kg body weight:

- Quantity of Busulfan:

$$\frac{Y \text{ (kg)} \times D \text{ (mg/kg)}}{6 \text{ (mg/ml)}} = A \text{ ml of Busulfan to be diluted}$$

Y: body weight of the patient in kg

D: dose of Busulfan

- Quantity of diluent:

$(A \text{ ml Busulfan}) \times (10) = B \text{ ml of diluent}$

To prepare the final solution for infusion, add (A) ml of Busulfan to (B) ml of diluent (sodium chloride 9 mg/ml (0.9%) solution for injection or glucose solution for injection 5%)

Preparation of the solution for infusion

- Busulfan must be prepared by healthcare professional using sterile transfer techniques. Using a non polycarbonate syringe fitted with a needle:
 - The calculated volume of Busulfan must be removed from the vial.
 - The contents of the syringe must be dispensed into an intravenous bag (or syringe) which already contains the calculated amount of the selected diluent. Busulfan must always be added to the diluent, not the diluent to the Busulfan. Busulfan must not be put into an intravenous bag that does not contain sodium chloride 9 mg/ml (0.9%) solution for injection or glucose solution for injection 5%.
- The diluted solution must be mixed thoroughly by inverting several times.

After dilution, 1 ml of solution for infusion contains 0.5 mg of Busulfan.

Diluted Busulfan is a clear colourless solution.

Instructions for use

Prior to and following each infusion, flush the indwelling catheter line with approximately 5 ml of sodium chloride 9 mg/ml (0.9%) solution for injection or glucose (5%) solution for injection.

The residual medicinal product must not be flushed in the administration tubing as rapid infusion of Busulfan has not been tested and is not recommended.

The entire prescribed Busulfan dose should be delivered over two hours.

Small volumes may be administered over 2 hours using electric syringes. In this case infusion sets with minimal priming space should be used (i.e 0.3-0.6 ml), primed with medicinal product solution prior to beginning the actual Busulfan infusion and then flushed with sodium chloride 9 mg/ml (0.9%) solution for injection or glucose (5%) solution for injection.

Busulfan must not be infused concomitantly with another intravenous solution.

Polycarbonate syringes must not be used with Busulfan.

For single use only. Only a clear solution without any particles should be used.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements for cytotoxic medicinal products.

OVERDOSE:

The principal toxic effect is profound myeloablation and pancytopenia but the central nervous system, liver, lungs, and gastrointestinal tract may also be affected.

There is no known antidote to Busulfan other than haematopoietic progenitor cell transplantation. In the absence of haematopoietic progenitor cell transplantation, the recommended dose of Busulfan would constitute an overdose of Busulfan. The haematologic status should be closely monitored and vigorous supportive measures instituted as medically indicated.

There have been two reports that Busulfan is dialyzable, thus dialysis should be considered in the case of an overdose. Since, Busulfan is metabolized through conjugation with glutathione, administration of glutathione might be considered.

It must be considered that overdose of Busulfan will also increase exposure to DMA. In human the principal toxic effects were hepatotoxicity and central nervous system (CNS) effects. CNS changes precede any of the more severe side effects. No specific antidote for DMA overdose is known.

In case of overdose, management would include general supportive care.

STORAGE:

Store in a refrigerator (2°C-8°C).
Do not freeze the diluted solution.

Diluted solution

- Busulfan Kabi Injection diluted in 0.9% Sodium Chloride Injection, or 5% Dextrose Injection, is stable at room temperature (25°C) for up to 8 hours but the infusion must be completed within that time.
- Busulfan Kabi Injection diluted in 0.9% Sodium Chloride Injection, is stable at refrigerated conditions (2°C to 8°C) for up to 12 hours but the infusion must be completed within that time.

From a microbiological point of view, the product should be used immediately after dilution. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than the above-mentioned conditions when dilution has taken place in controlled and validated aseptic conditions.

INCOMPATIBILITIES:

Busulfan must be diluted prior to use with either 0.9% Sodium Chloride Injection (normal saline) or 5% Dextrose Injection. Busulfan should not be mixed with other medicinal products.

Polycarbonate syringes must not be used with Busulfan.

PRESENTATION:

Busulfan Kabi Injection 6 mg/ml Concentrate for Solution for Infusion is packaged as a sterile solution in 10 ml single-use clear glass vials each containing 60 mg of busulfan at a concentration of 6 mg/ml for intravenous use.

Pack size: 1 x 10 ml vial, 8 x 10 ml vials

MANUFACTURER INFORMATION:

Manufactured in India by:
Fresenius Kabi Oncology Ltd.

Village Kishanpura,
Baddi,
Tehsil Nalagarh,
District Solan,
Himachal Pradesh,
174101 INDIA

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